

Program -3

Code a Dockerized Python Flask or Node.js Application

DevOps

- pgm3
- Dockerfile
- app.py
- requirements.txt

Step 1:

`mkdir pgm3`

Creates a new folder to store all files related to Program-3.

`cd pgm3`

Enters the newly created folder to work inside it.

`nano Dockerfile`

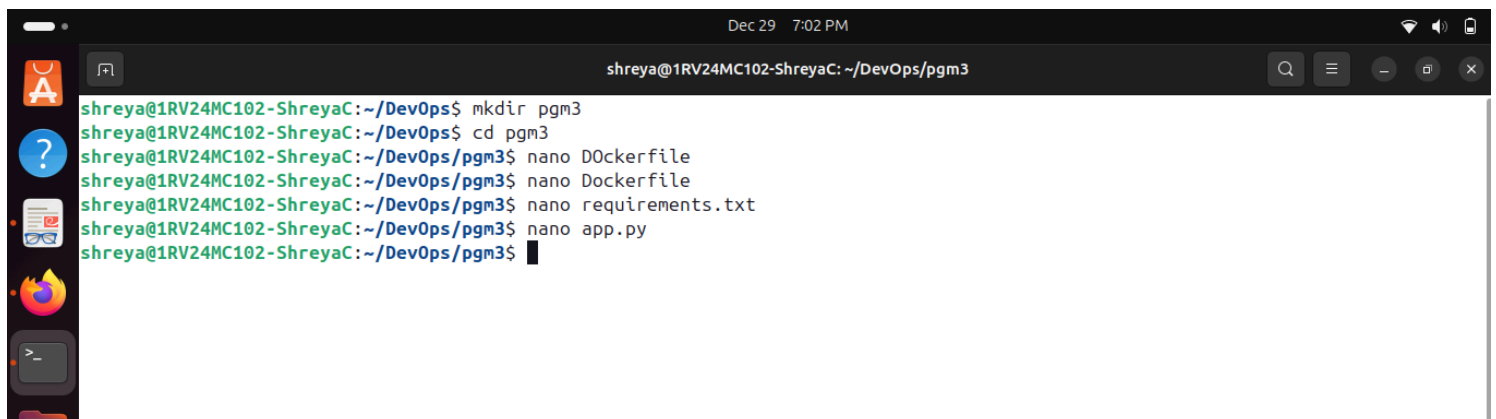
Opens the Dockerfile to define how the application image is built.

`nano requirements.txt`

Opens the file to list required Python packages for the application.

`nano app.py`

Opens the Python file to write the Flask web application code.



```
Dec 29 7:02 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/pgm3

shreya@1RV24MC102-ShreyaC:~/DevOps$ mkdir pgm3
shreya@1RV24MC102-ShreyaC:~/DevOps$ cd pgm3
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ nano Dockerfile
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ nano Dockerfile
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ nano requirements.txt
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ nano app.py
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$
```

Step 2 : Add the following code in Dockerfile



The screenshot shows a terminal window with the nano text editor open to a file named 'Dockerfile'. The editor's title bar indicates the user is 'shreya@1RV24MC102-ShreyaC' in the directory '~/DevOps/pgm3'. The file content is as follows:

```
GNU nano 7.2 Dockerfile
# Use official Python image
FROM python:latest

# Set working directory
WORKDIR /app

# Copy requirement file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY . .

# Expose port 5000
EXPOSE 5000

# Run the app
CMD ["python", "app.py"]
```

The bottom of the terminal shows the nano editor's command shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, M-U Undo, M-A Set Mark, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, ^_ Go To Line, M-E Redo, and M-6 Copy. A status bar at the bottom indicates '[Read 18 lines]'.

Step 3 : Add the following code in [app.py](#)



The screenshot shows a terminal window with the nano text editor open to a file named 'app.py'. The editor's title bar indicates the user is 'shreya@1RV24MC102-ShreyaC' in the directory '~/DevOps/pgm3'. The file content is as follows:

```
GNU nano 7.2 app.py
from flask import Flask

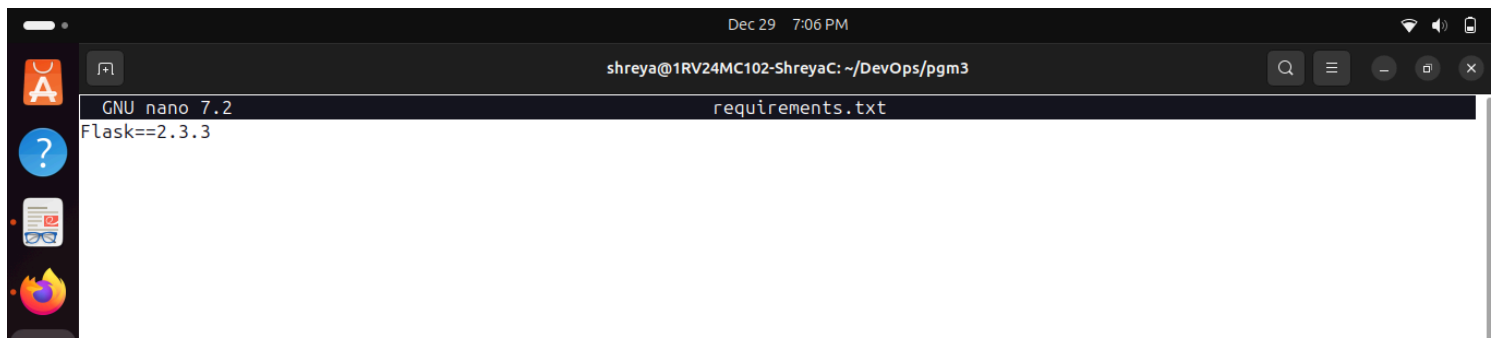
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

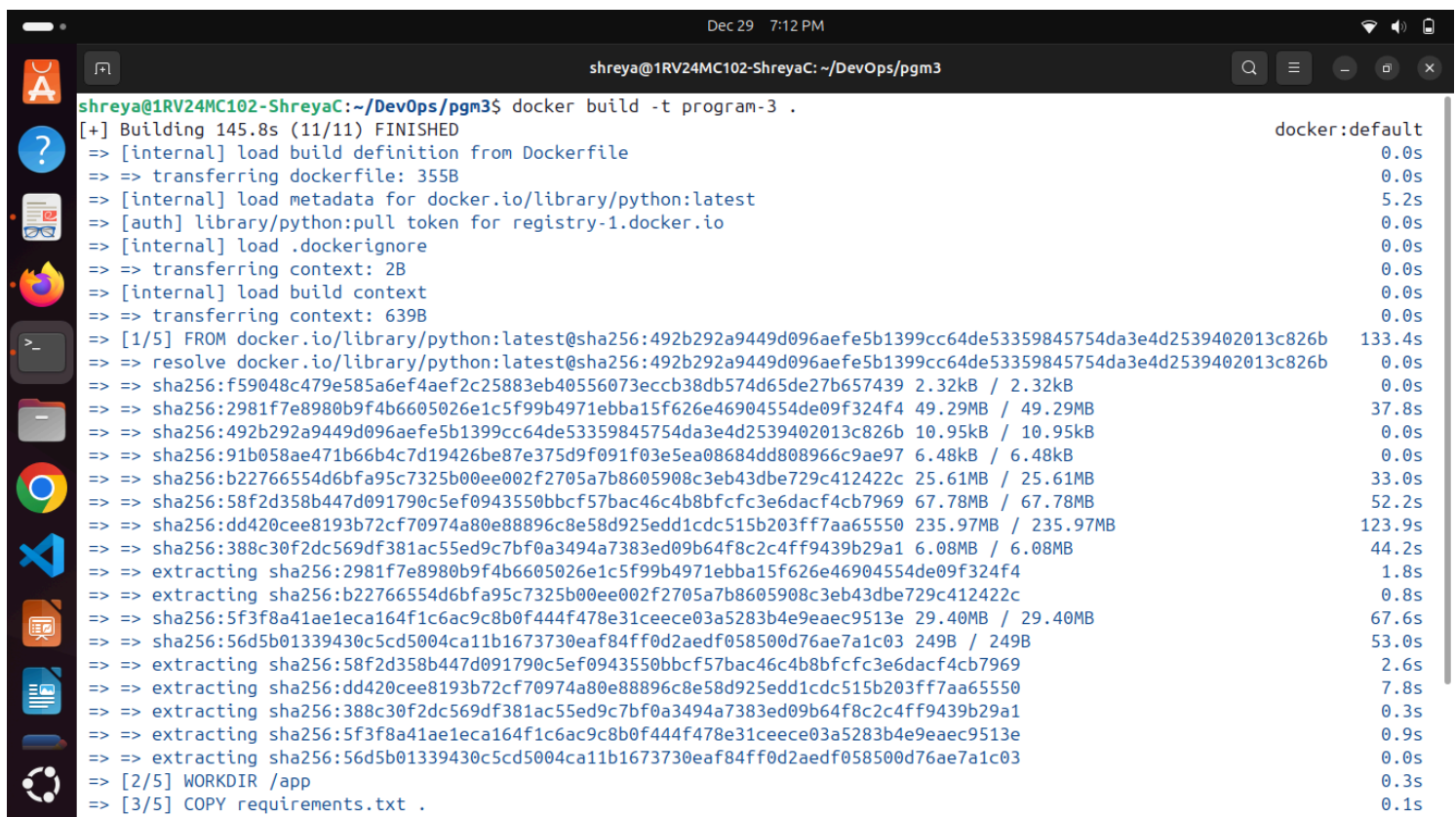
The bottom of the terminal shows the nano editor's command shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, M-U Undo, M-A Set Mark, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, ^_ Go To Line, M-E Redo, and M-6 Copy. A status bar at the bottom indicates '[Read 11 lines]'.

Step 4 : Add the following code in requirements.txt file



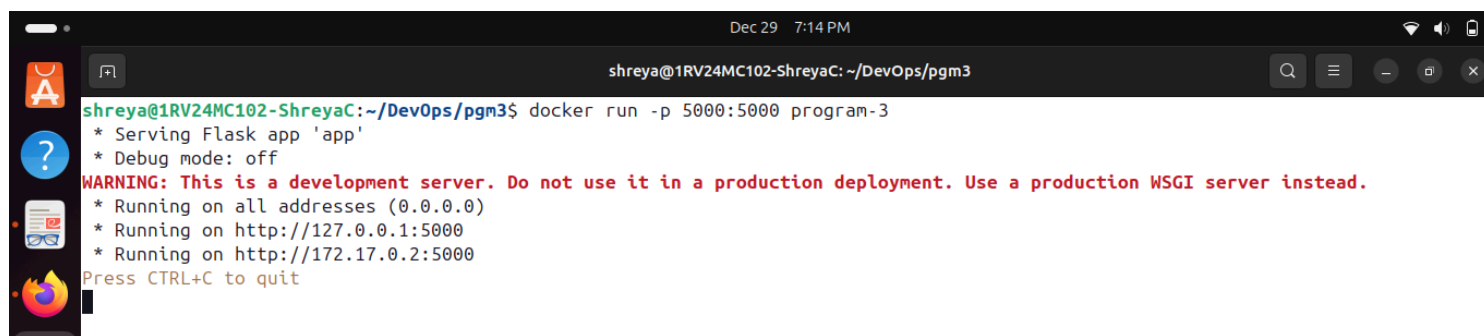
```
Dec 29 7:06 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/pgm3
GNU nano 7.2 requirements.txt
Flask==2.3.3
```

Step 5 : Build a Docker image named program-3 using the Dockerfile in the current directory.



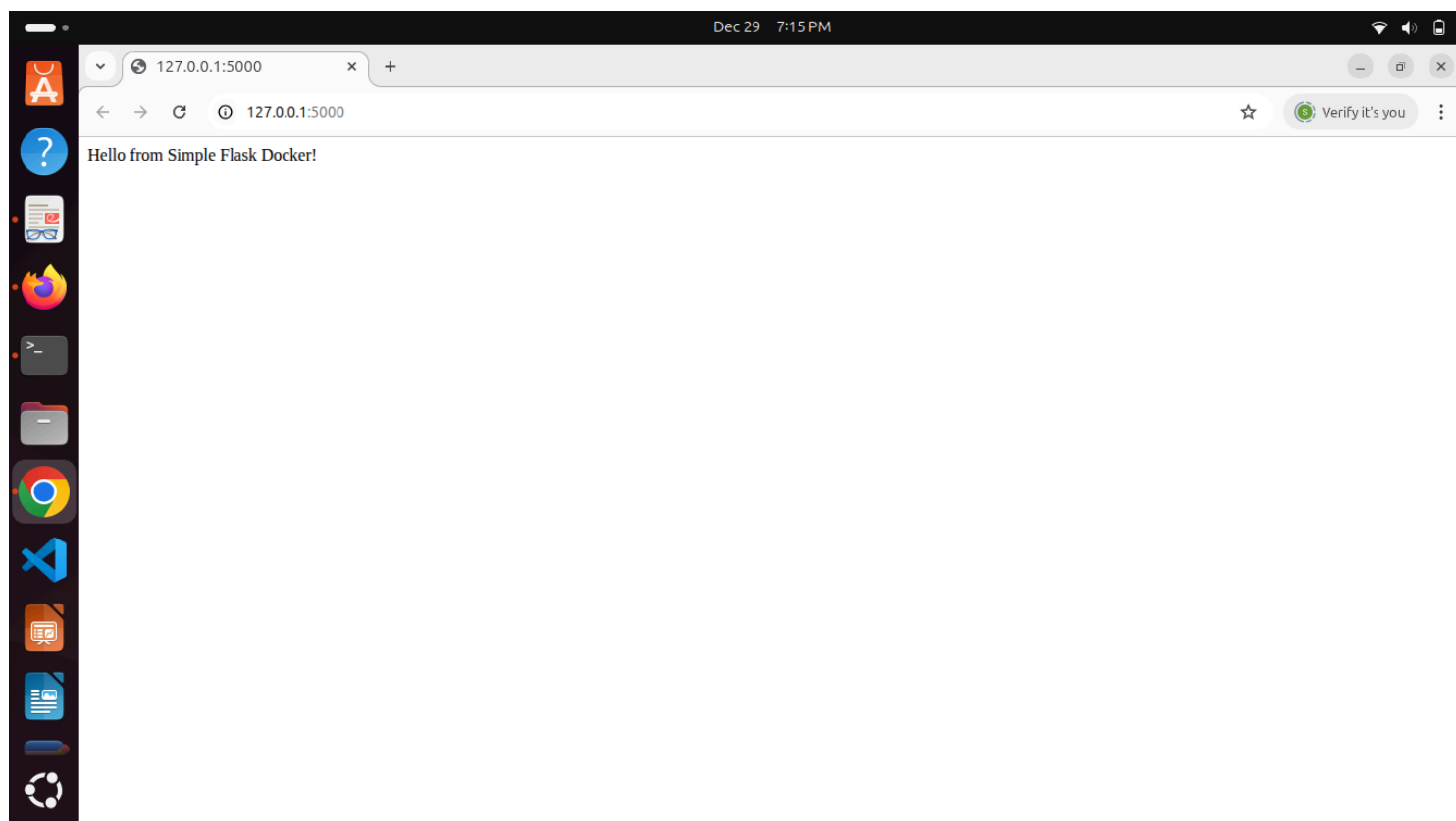
```
Dec 29 7:12 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/pgm3$ docker build -t program-3 .
[+] Building 145.8s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 355B
=> [internal] load metadata for docker.io/library/python:latest
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 639B
=> [1/5] FROM docker.io/library/python:latest@sha256:492b292a9449d096aefe5b1399cc64de53359845754da3e4d2539402013c826b
=> => resolve docker.io/library/python:latest@sha256:492b292a9449d096aefe5b1399cc64de53359845754da3e4d2539402013c826b
=> => sha256:f59048c479e585a6ef4aef2c25883eb40556073eccb38db574d65de27b657439 2.32kB / 2.32kB
=> => sha256:2981f7e8980b9f4b6605026e1c5f99b4971ebba15f626e46904554de09f324f4 49.29MB / 49.29MB
=> => sha256:492b292a9449d096aefe5b1399cc64de53359845754da3e4d2539402013c826b 10.95kB / 10.95kB
=> => sha256:91b058ae471b66b4c7d19426be87e375d9f091f03e5ea08684dd808966c9ae97 6.48kB / 6.48kB
=> => sha256:b22766554d6bfa95c7325b00ee002f2705a7b8605908c3eb43dbe729c412422c 25.61MB / 25.61MB
=> => sha256:58f2d358b447d091790c5ef0943550bbcf57bac46c4b8bfcfc3e6dacf4cb7969 67.78MB / 67.78MB
=> => sha256:dd420cee8193b72cf70974a80e88896c8e58d925edd1cdc515b203ff7aa65550 235.97MB / 235.97MB
=> => sha256:388c30f2dc569df381ac55ed9c7bf0a3494a7383ed09b64f8c2c4ff9439b29a1 6.08MB / 6.08MB
=> => extracting sha256:2981f7e8980b9f4b6605026e1c5f99b4971ebba15f626e46904554de09f324f4 1.8s
=> => extracting sha256:b22766554d6bfa95c7325b00ee002f2705a7b8605908c3eb43dbe729c412422c 0.8s
=> => sha256:5f3f8a41ae1eca164f1c6ac9c8b0f444f478e31ceece03a5283b4e9aeca9513e 29.40MB / 29.40MB
=> => sha256:56d5b01339430c5cd5004ca11b1673730eaf84ff0d2aedf058500d76ae7a1c03 249B / 249B
=> => extracting sha256:58f2d358b447d091790c5ef0943550bbcf57bac46c4b8bfcfc3e6dacf4cb7969 2.6s
=> => extracting sha256:dd420cee8193b72cf70974a80e88896c8e58d925edd1cdc515b203ff7aa65550 7.8s
=> => extracting sha256:388c30f2dc569df381ac55ed9c7bf0a3494a7383ed09b64f8c2c4ff9439b29a1 0.3s
=> => extracting sha256:5f3f8a41ae1eca164f1c6ac9c8b0f444f478e31ceece03a5283b4e9aeca9513e 0.9s
=> => extracting sha256:56d5b01339430c5cd5004ca11b1673730eaf84ff0d2aedf058500d76ae7a1c03 0.0s
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install -r requirements.txt
=> [5/5] COMMIT program-3
```

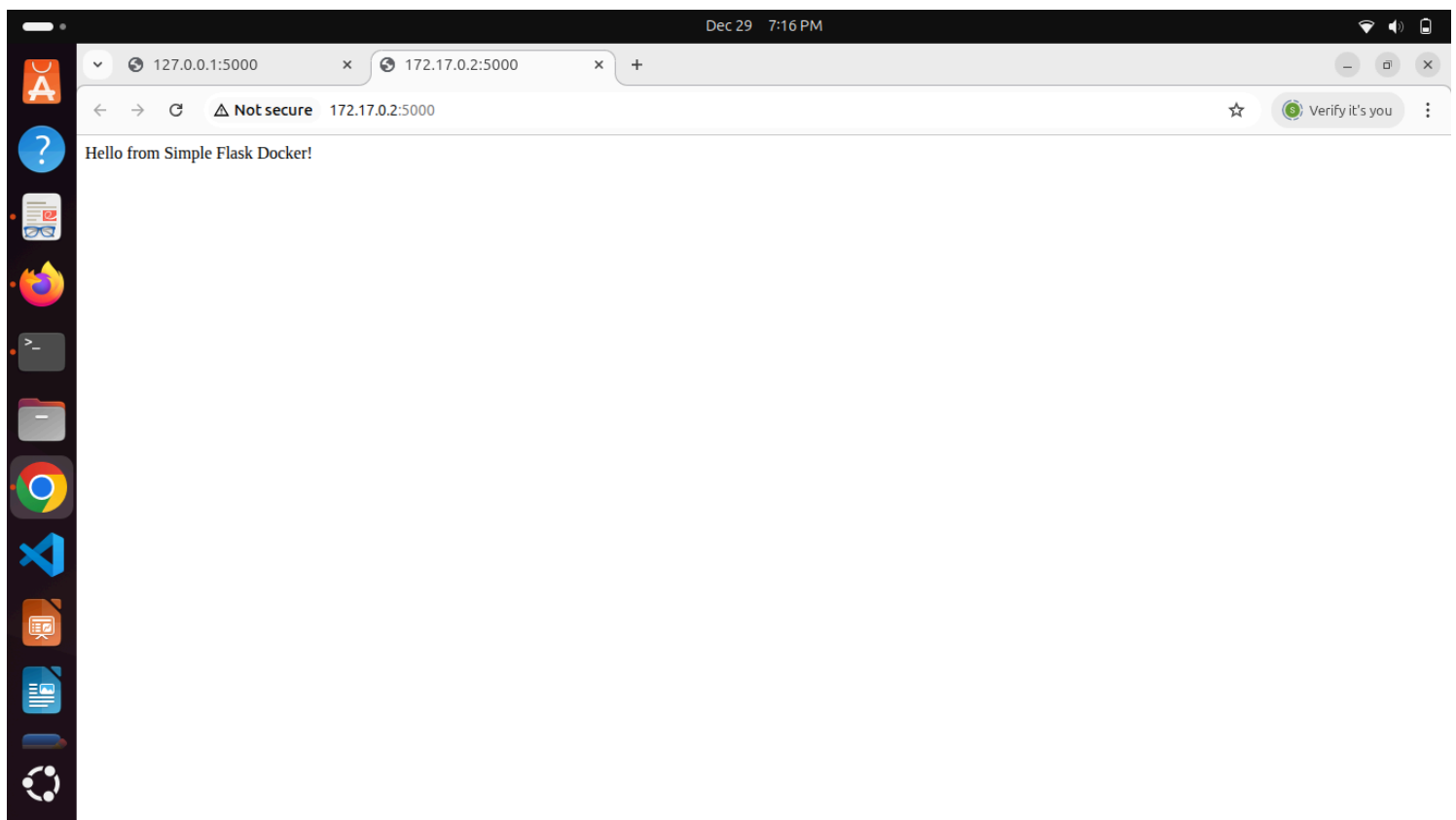
Step 6 : Runs the Docker container from the program-3 image and maps container port 5000 to host port 5000 so the app can be accessed in the browser.



A terminal window titled "Dec 29 7:14 PM" showing a user at "shreya@1RV24MC102-ShreyaC" in the directory "~/DevOps/pgm3". The user has executed the command "docker run -p 5000:5000 program-3". The output shows the Flask application starting, serving on all addresses (0.0.0.0) and specifically on http://127.0.0.1:5000 and http://172.17.0.2:5000. A warning message is displayed: "WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead." The prompt "Press CTRL+C to quit" is visible at the bottom.

```
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ docker run -p 5000:5000 program-3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
```





Step 7 :

`docker container ls -a`

Displays all Docker containers, including running and stopped ones.

`docker container stop <container-number>`

Stops the specified running Docker container.

`docker container rm <container-number>`

Removes (deletes) the specified Docker container.

`docker image rm program-3`

Deletes the Docker image named program-3 from the system.

```
Dec 29 7:30 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/pgm3
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ docker container stop a7987f9dab02
a7987f9dab02
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ docker container rm a7987f9dab02
a7987f9dab02
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ docker image rm program-3
Error response from daemon: conflict: unable to remove repository reference "program-3" (must force) - container 8ada689c7add is
using its referenced image f0cefdeffb3d
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$ docker image rm -f program-3
Untagged: program-3:latest
Deleted: sha256:f0cefdeffb3d71f6f91c2d81d57220dcb799de0c33c274063598bc2e5b28f85a
shreya@1RV24MC102-ShreyaC:~/DevOps/pgm3$
```